Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/30/2015
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Project Title:	Effects of Stem Cell Morphology and Motility on Osteogenesis in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2015 per A. Chu/ARC and NSSC information (Ed., 8/5/14) NOTE: To be extended, per PI (Ed., 5/28/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Yew, Alvin (NASA Goddard Space Flight Center)		
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Task Description:	Mesenchymal stem cells (MSCs) are responsible for tissue development, help maintain homeostasis in mature tissues, and are being considered for therapeutic treatments of a variety of cellular lineages including bone, skeletal muscles, and fatty tissues. Chemical and mechanical factors modulate lineage commitment; however, spaceflight can drastically change these cues and hence, influence physiologic processes, such as bone formation. We hypothesize that such changes arise from altered cell-cell interactions and chemotactic behaviors brought about by cell morphologies and cytoarchitectures adopted during microgravity. We seek to investigate the effects of microgravity on MSC morphology, migration, and osteogenic differentiation using a newly developed clinorotation time-lapse microscopy (CTM) system and lab-on-chip devices to generate chemical gradients. While the underlying principles of CTM are identical to conventional clinostats, CTM is amenable to live-cell imaging, without prohibitively large experimental setups or disruption of culture environments. Importantly, CTM represents a significant step forward in space biology research, because it enables ground-based microgravity studies of not only traditional end-point outcomes, but also dynamic cellular processes. Specifically, we aim to: (1) investigate the long-term, dynamic behaviors of MSCs under chemokine gradients in simulated microgravity by analyzing the time-dependent alterations to morphology, motility, and cell density and (2) quantify changes in distributions of focal adhesions and osteogenic phenotype indicators by using CTM to monitor a live-cell alkaline phosphatase (AP) stain and end-point vinculin, lipids, and AP stains. This investigation provides novel insight into how dynamic MSC events are affected by microgravity. We believe that our hypothesis may partially explain a mechanism responsible for reduced bone mass and impaired fracture healing in space. Spinoff research includes: investigating genomic and proteomic profiles associated with MSC cytoarchitecture and differentiation; assessing the potential of MSCs in treating musculoskeletal pathologies in spaceflight; using drugs to promote normal MSC-mediated morphogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Spinoff research includes: investigating genomic and proteomic profiles associated with MSC cytoarchitecture and differentiation; assessing the potential of MSCs in treating musculoskeletal pathologies in spaceflight; using drugs to promote normal MSC-mediated morphogenesis.
Task Progress:	Under this NASA grant (NNX13AM06G), we have been able to perform unprecedented experiments toward elucidation of the biophysical regulation of human mesenchymal stem cells (MSCs) in simulated microgravity. Using our one-of-a-kind clinochip-platform, we performed clinorotation time-lapse microscopy to visualize the dynamic behavior of cells under simulated microgravity conditions. Our aims were to (1) investigate the dynamic time-dependent alterations to morphology, motility, and density of MSCs subjected to long-term simulated microgravity in chemokine gradients, and (2) quantify changes in focal adhesion distributions and osteogenic phenotype indicators. In our first set of experiments using the clinochip system, we demonstrated that MSC spreading events are profoundly affected by the angular frequency of clinorotation. In contrast to our expectations, MSC motility appeared to be influenced more by cell shape than by clinorotation directly. Regardless of rotation speed, circular MSCs exhibited little migration, while elongated MSCs migrated similarly to those in standard gravity. Because cell shape was the most sensitive to clinorotation, we expanded the scope of Aim 1 to include the development of a technique to study cell-substrate interactions. This led to our first ever characterization of cell traction forces during clinorotation. Toward the completion of Aim 2, we worked in parallel to build a new system for long-term clinorotation of cells in clinochips. Our findings of cell rounding in clinorotated cells suggested that simulated microgravity might have an adverse effect on osteogenesis, as others have shown using soft or micropatterned islands to limit cell attachment. Consistent with these previous reports, our MSCs did not exhibit any osteogenesis when subjected to clinorotation over 14 days, while cells in standard gravity were robustly osteogenic. Overall, our work under this NASA grant has laid the foundation toward future studies in the characterization of molecular mechanisms in gravisensing and in the discovery of therapeutic countermeasures against adverse health changes during space exploration. Our experiments have revealed that clinorotation speed is an important parameter for MSC perception of microgravity under dynamic conditions, but may be less significant for extended durations of microgravity exposure. In particular, there appears to be an inversely graded response of the degree of MSC spreading with angular frequency. We hypothesize that this may be related to differences in characteristic times of different molecular processes that are involved in cell spreading. Since we observed no angular frequency with cell retraction, this hypothesis implies that other molecular processes might not be similarly regulated. Interestingly, over the long-term, the differential effects of angular frequency on cell morphology appear to converge toward that observed at higher rotational speeds (i.e., 75 rpm). These results support our hypothesis that the lower rotational speed of 30 rpm is too slow for time-averaging within the characteristic times for processes involved in cell spreading. However, with prolonged exposure, these processes will statistically have greater likelihood of becoming nullified at some point in time. This being the case, we found that cell areas did not tell the entire story. Analysis of adhesion areas using surface reflective interference contrast microscopy revealed differences between 30 rpm and 75 rpm long-term clinorotation, even while cell areas were not drastically different. This also supports the hypothesis that multiple mechanisms may govern cell-substrate interactions. The differences in morphologies of MSCs exposed to clinorotation suggests a fundamental change in the way cells interact with their surrounding matrix environment. To characterize this, we performed the first ever demonstration of cell traction microscopy during clinorotation and showed that both magnitude and distribution of forces exerted by MSCs are altered. These data indicate that simulated microgravity has a profound biophysical influence on how MSCs interact with their surrounding microenvironment. Lastly, and most dramatically, we show that long-term clinorotation exposure results in the inability for MSCs to differentiate into bone cells, even under biochemically permissive environments. The profound effect of clinorotation on the ability for MSCs to differentiate into bone has potentially transformative impact on our understanding of bone remodeling during long-term exposure to microgravity environments. Since MSCs are required to infiltrate regions of resorbed bone to give rise to the osteoblasts that form new bone, an inability to differentiate effectively obstructs normal bone turnover.

Bibliography Type:	Description: (Last Updated: 10/30/2015)
Abstracts for Journals and Proceedings	Hsieh AH, Luna C, Yew AG "Cellular gravisensing thresholds with clinorotation." Animals – Ground Studies I. Presented at 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Luna C, Falohun T, Yew AG, Hsieh AH. "Biophysical aspects of osteosarcoma cells in simulated microgravity: cell-substrate adhesion, migration and cell-cell interactions." Humans #1 (Cell Based Studies). Presented at 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Yew AG, Hsieh AH. "Frequency response of stem cells to gravity loads." Poster session. Presented at the 7th World Congress of Biomechanics, Boston, MA, July 11-14, 2014. 7th World Congress of Biomechanics, Boston, MA, July 11-14, 2014. , Jul-2014
Abstracts for Journals and Proceedings	Luna C, Yew AG, Hsieh AH. "Forces behind cell adhesion and migration in microgravity." Poster session. Presented at the 59th Annual Meeting of the Biophysical Society, Baltimore, MD, February 7-11, 2015. Biophysical Journal. 2015 Jan 27;108(2 Suppl 1):457a. http://dx.doi.org/10.1016/j.bpj.2014.11.2495 , Jan-2015
Abstracts for Journals and Proceedings	Luna C, Stevick RJ, Yew AG, Hsieh AH. "Relationship between cell-adhesion, matrix elasticity, traction forces and the response of mesenchymal stem cells to simulated microgravity." Investigator Poster Session. Presented at 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Articles in Peer-reviewed Journals	Luna C, Yew AG, Hsieh AH. "Effects of angular frequency during clinorotation on mesenchymal stem cell morphology and migration." npj Microgravity. 2015 Jul 30;1:15007. http://dx.doi.org/10.1038/npjmgrav.2015.7 , Jul-2015